DSGE Model Estimation via VAR IRF Matching: The Short and Long of It *

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Abstract

Following De Graeve and Westermark (2025) and Baumeister (2025), I increase the lag length of the VAR in Altig et al. (2005) to re-estimate their DSGE. Moving from $p = 4$ to $p = 12$ lags, I find that both the VAR and DSGE impulse responses, and the implied structural parameters, change substantially. The Calvo price parameters that distinguish the observationally equivalent firm-specific and homogeneous capital versions of the model converge to one another, from 0.34 against 0.82 at $p = 4$ to 0.89 against 0.90 at $p = 12$. The implied duration of price contracts in the firm-specific model rises from 1.5 to 9.0 quarters, equal to that of the homogeneous capital model, re-opening the micro–macro discrepancy in price stickiness that firm-specific capital was introduced to resolve.

Keywords: Truncation Bias, Long-Lag VARs, DSGE Estimation, IRF Minimum Distance , Nominal Rigidities
JEL Codes: C32, C52, E32

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1 Introduction

1.1 Motivation

Inflation is persistent; when it appears is typically going to stay for some time, and it is difficult to bring down. This is a stylised fact of the data, as can be seen in the post second world war US inflation history in Figure 1.

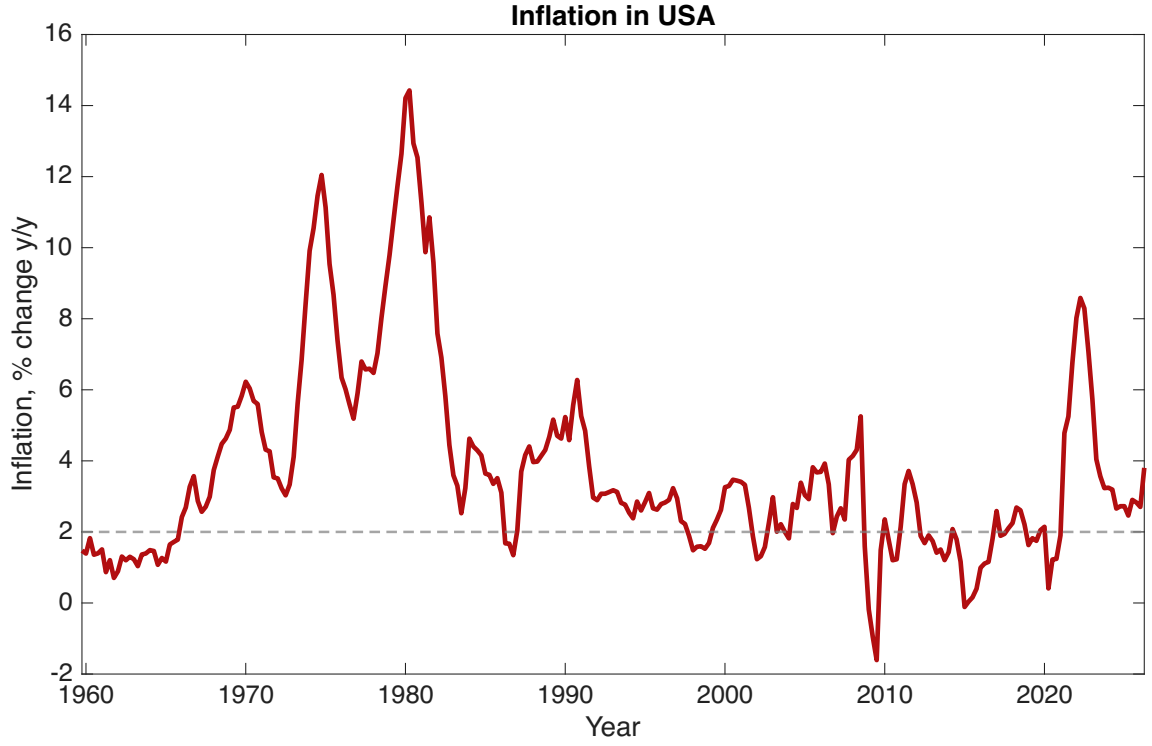


Figure 1: Inflation in the US, 1959Q2–2026Q2: quarterly growth rate of the Consumer Price Index, in percent.

Many Dynamic Stochastic General Equilibrium (DSGE) models are built to explain this phenomena. But standard New Keynesian (NK) models are able to model this inflation behaviour only impling inplausibly high levels of price stickiness, which are in stark contrast with micro-data evidence. Altig et al. (2011) propose a resolution to this micro-macro conflict. They create two differend models built on two versions of the Christiano et al. (2005) model: one with *homogeneous* capital and the other were the capital is *firm-specific*.

One of the important goals of monetary business cycle research is to explain the empirical evidence on the impact of economic shocks on macroeconomic variables. Because the systematic component of policy is endogenous, identifying the causal effect of monetary policy shocks requires isolating the non-systematic component, the monetary policy shock, even though it is small and infrequent Ravenna (2007); Aruoba and Drechsel (2024).

A large class of estimated DSGE models is taken to the data through this channel. Impulse Response Function Matching Estimation (IRFME) is a limited-information strategy that chooses structural parameters to minimise the distance between the impulse responses implied by the model and those estimated from an identified VAR (Christiano et al., 2005; Altig et al., 2011; Rossi et al., 2012). The strategy is attractive precisely because it is partial: the researcher takes a stand on the identification of a subset of shocks, and remains agnostic about the remaining dimensions of the data.

That attractiveness rests entirely on one maintained assumption: that the VAR delivers *consistent* estimates of the moments being matched. If the reduced-form VAR adequately captures the data generating process, impulse responses are optimal at all horizons; if the VAR is misspecified, the specification error propagates to the structural statistics and compounds at each horizon (Ramey, 2016; Canova and Fosso, 2025) Ravenna (2007) "The bias introduced by truncation can lead to bias in the identification of the structural shock." Altig et al. (2011, footnote 10, p. 3) say as much themselves: "the quality of our estimation strategy depends on the ability of identified VARs to generate reliable estimates of the dynamic response of economic variables to shocks".

Since a VAR(∞) cannot be estimated on a finite sample, researchers truncate. The resulting VAR(p) is misspecified, and increasingly so the smaller is p (De Graeve and Westermark, 2025; Barnichon et al., 2026; Ravenna, 2007). Ravenna (2007) "the problem facing economistis is whether the number of lags p to be included is reasonable given the lenght of economics time series over which VARs are estimated. When estimating a VAR consistent with DSGE business cycle models it is standard to assume that including few lags is sufficient to provide a reasonable approximation to the exact infinite order VAR representation! De Graeve and Westermark (2025) show that the conventional objection to longer lags, that the additional parametrisation makes confidence intervals prohibitively wide, is incomplete, because reducing misspecification reduces not only bias but also variance. Their conclusion is addressed explicitly to applied researchers: "it is up to the researcher to show that her structural results are robust to significant increases in the lag length, far beyond the lag lengths typically considered in current practice".

1.2 Contribution and Results

This paper asks what lag truncation does to *structural* estimates, not merely to impulse responses. I replicate Altig et al. (2011) — henceforth ACEL — and re-estimate their DSGE by IRF matching against VARs of increasing lag order.

The contribution is twofold. First, relative to De Graeve and Westermark (2025), I move the question from the reduced form to the structural parameters. Second, and

more specifically, I do so in a setting their design deliberately excludes. De Graeve and Westermark (2025) state that "all our experiments are based on two-shock models and two-variable VARs", and that where a candidate model does not fit that framework — Smets and Wouters — "we consider the model with only monetary policy and preference shocks".[1] Their data generating processes are square and fundamental by construction, so that truncation bias is isolated from singularity and invertibility. ACEL's model is not: it confronts three structural shocks with a ten-variable VAR. [**TBD:** Verificare che nessun altro abbia gia' fatto questo esercizio su un modello stocasticamente singolare. Se confermato, e' il claim centrale.]

My main results are the following. *First*, the impulse responses to a monetary policy shock retain their sign and broad shape across lag orders, but the humps peak substantially later as p increases, and the point estimates remain inside the VAR(4) confidence bands. *Second*, the responses to the two technology shocks — identified with long-run rather than short-run restrictions — do not: several fall outside the baseline confidence set. This asymmetry is what Ravenna (2007) predicts, since long-run restrictions require the sum of the entire lag polynomial and truncation error accumulates in exactly that object. *Third*, the structural estimates move. The Phillips curve slope falls from 0.0402 to 0.0129 and becomes precisely estimated only at $p = 12$; the cost of varying capital utilisation collapses from 1.99 to 0.004; the investment adjustment cost roughly doubles. *Fourth*, and as a consequence, the implied Calvo price parameters of the firm-specific and homogeneous capital models converge, from 0.34 against 0.82 to 0.89 against 0.90. The implied duration of price contracts in the firm-specific model rises from 1.5 to 9.0 quarters.

The last result overturns ACEL's central conclusion. Their claim is that only the firm-specific capital model reconciles macro evidence on inflation inertia with micro evidence on the frequency of price adjustment. Under a richer lag structure that reconciliation disappears. The channel is transparent and internal to their own argument: ACEL note that "to the extent that capital utilization rates can easily be varied, the assumption of firm-specific capital loses its ability to rationalize low values of ξ_p".[2] Long-lag VARs push the estimate of σ_a precisely into that region.

1.3 Related Literature

Long-lag VARs.
Truncation bias and the DSGE–VAR mapping.
Impulse response matching and DSGE estimation.

[1] (De Graeve and Westermark, 2025, p. 10). They also note that "invertibility is never a problem; all the models we consider are fundamental".
[2] (Altig et al., 2011, p. 227).

1.4 Outline

Section

2 The Theoretical Model

2.1 Inflation inertia and the micro–macro conflict

Altig et al. (2011) start from a tension in the New Keynesian literature. To account for inertia in inflation, standard models require firms to re-optimise prices infrequently; every six quarters, and in versions without indexation to past inflation even every two years. This sits uneasily with the microeconometric evidence, which finds price changes considerably more frequent (Bils and Klenow, 2004; Golosov and Lucas, 2007; Klenow and Kryvtsov, 2008).

Altig et al. (2005)[3] propose a resolution. In their model inflation is inertial even though firms re-optimise on average every 1.5 quarters. Inertia arises not because firms change prices rarely, but because when they do they change them by a small amount: each firm's short-run marginal cost is increasing in its own output, which dampens the optimal price adjustment. The underlying assumption that drives this result is that capital is predetermined within the period and firm-specific, so that factors of production cannot be instantly and costlessly reallocated across firms.

2.2 Homogeneous and firm-specific capital

Altig et al. (2011) conduct the analysis using two versions of the Christiano et al. (2005) model: one in which capital is homogeneous, as in standard formulations, and the other in which it is firm-specific.

Once log-linearised, the only difference between the two sets of equilibrium conditions lies in the equation relating inflation to marginal cost, the New Keynesian Phillisp Curve,

$$\Delta\hat{\pi}_t = \mathbb{E}\big[\beta\Delta\hat{\pi}_{t+1} + \gamma\hat{s}_t \mid \Omega_t\big], \tag{1}$$

where $\hat{s}_t$ is economy-wide average real marginal cost in units of the final good, and the slope γ maps into the structural parameters as

$$\gamma = \frac{(1-\xi_p)(1-\beta\xi_p)}{\xi_p}\,\chi. \tag{2}$$

The information set Ω_t contains the current realisation of the technology shocks but not that of the monetary policy innovation. The two models are parametrised in terms of γ rather than ξ_p.

Given the remaining parameters they are then observationally equivalent with respect

[3]The replicaiton code made available from the authors is NBER working paper from 2005. Hence, I replicate and refere to those results.

to aggregate prices and quantities, so that no stand need be taken on which version is being estimated. What distinguishes them is the value of ξ_p given γ implies.

The real rigidity parameter χ measures the strategic complementarity in price setting of Ball and Romer (1990). It takes values in $(0, 1]$, and the closer it lies to zero the stronger the attenuation. That is, a firm that is able to reset its price moves it by less, for a given movement in economy-wide marginal cost. In the *homogeneous* capital model χ is set to 1, the attenuation is absent. In the *firm-specific* model χ is a non-linear function of the structural parameters.

In particular χ depends on some calibrated parameters, $[a, \beta, \delta, \lambda_f]$ which are respectively, capital component in the Cobb-Douglas production function, time preference parameter, capital depreciation rate, and the average price markup. And two estimated parameters, $[\sigma_a; S'']$, which are the Capacity utilisation cost and Investment adjustment cost.

The Altig et al. (2011) logic suggest that, for a fixed value of ξ_p, the larger σ_a, the lower is γ, in line with the vast set of literature that suggests that the Phillips curve slope is low. But other things equal, a lower ξ_p implies a higher γ. This determine that for a given value of γ , ξ_p is a decreasing function of σ_a.
In this sense, a flat Phillips curve is a problem in the homogenoues capital model. A low value of γ implies that, given that there $\chi = 1$, the Clavo price stickiness parameter ξ_p need to be very high, implying irrealistic implied time duration of prices, that is $D = \frac{1}{(1-\xi_p)}$ which are in stark conflict with micro evidence.

The role of σ_a is easiest to see by following what happens after an unexpected, positive demand shock. Aggregate demand rises within the quarter and firms have to produce more. Hours expand first, and wages are themselves set on a Calvo scheme, so only the $1 - \xi_w$ of households that can reoptimise raise their wage while the remaining ξ_w keep supplying at the wage set earlier. The aggregate nominal wage therefore rises, and the real wage with it, but by little. Capital is another matter, and it is here that the two models differentiate from each other.

Under homogeneous capital there is a competitive, economy-wide market for capital services. The aggregate capital stock is predetermined within the quarter, so the additional services have to come from higher utilisation, and the rental rate rises by an amount governed by σ_a. The individual firm, however, takes that rental rate as given and can hire as many capital services as it wants at it. This is the standard assumption in equilibrium business cycle models, and Altig et al. (2005) state it plainly: "a firm's capital stock is not pre-determined and all factors of production, including capital, can be instantaneously transferred across firms, without any cost, in perfectly competitive markets".[4] Both of

[4](Altig et al., 2005, p. 2).

its factors are therefore available at prices it does not influence, so its marginal cost is the same as every other firm's and does not depend on how much this particular firm happens to produce. Marginal cost is common and independent of own output, which is what $\chi = 1$ records.

Under firm-specific capital that market does not exist. The firm is left with the capital it already owns, which cannot be reallocated within the period, so the only way to obtain more capital services is to run its own stock harder. It would be willing to pay for additional services, but there is no market on which to buy them, and σ_a is what raising utilisation costs it instead. When σ_a is high that margin is expensive, and most of the additional output has to come from adding hours to a quantity of capital services that barely moves. The marginal product of labour falls as the fixed factor binds, and the firm's own marginal cost rises with its own output.

This changes the decision of the $1 - \xi_p$ firms that can reset their price. Economy-wide marginal cost has risen, so optimiser firms desire to raise prices. But a firm that raises its price by more than the others sees its relative price rise, its demand fall, and its own output fall with it, and under firm-specific capital a lower own output means a lower own marginal cost, which argues for a smaller increase in the first place. Having internalised that producing more is what raises its costs, the firm moves its price by less than it would in the absence of this real rigidity, the more so the steeper its own marginal cost curve and the more elastic the demand it faces, since a given price increase then costs it more output. Inflation responds less to a given movement in economy-wide marginal cost, and this is precisely what a χ below one means.

It is worth being precise about what is doing the work here, because σ_a is a single parameter, estimated once, which does not take different values across the two models. What differs is whether it reaches χ at all. Under homogeneous capital the market for capital services severs the link between a firm's own output and its own marginal cost, and $\chi = 1$ whatever σ_a happens to be. Under firm-specific capital the firm is tied to the capital it already owns, and σ_a decides how binding that tie is.

The limiting case makes the point. If σ_a is close to zero, utilisation is almost free to vary, so a firm tied to its own capital can still draw from it whatever volume of capital services it needs, at negligible cost. Owning the capital rather than renting it then makes no practical difference: the firm's own marginal cost is as flat as it would be with a market for capital services, χ converges to one, and the firm-specific model collapses onto the homogeneous one. Firm specificity is an assumption about ownership, but its economic content comes entirely from the cost of varying utilisation, and without that cost the constraint exists in name only.

This is where the mechanism is fragile, and Altig et al. (2011) say themselves so: *"to the extent that capital utilization rates can easily be varied, the assumption of firm-specific*

capital loses its ability to rationalize low values of ξ_p*"*.Altig et al. (2011, p. 227). The reconciliation of micro and macro evidence rests, in the end, lies on a single estimated parameter.

2.3 DSGE estimation via IRF matching

Theoretical models are usefull to economists and policymakers only if they are able to reproduce important feauters of the observed data. Which is a property that critically depends on the values that model's parameters have attached, and also on how the researcher decide to estimate them Theodoridis (2011).

Altig et al. (2005) folowing Rotemberg and Woodford (1997); Christiano et al. (2005); Iacoviello (2005); Boivin and Giannoni (2006), use a limited information approach the (IRFME), this procedure select and estimate the model's structural parameters that produce the DSGE IRFs, which minimises the distance with IRFs created from the SVAR.

Using an IRFME implies that the researcher is focusing to match on some limited aspects of the DSGE Giacomini (2013); it is choosing an identification strategy for the reduced form VAR in a way that is consistent with the key determinants of the DSGE. The VAR and theoretical IRFs are identified by restrictions implied by the DSGE. Which means that the DSGE admits a structural VAR representation, in this way, sample moments of the VAR are informative for the theoretical model's structural parameters Rossi et al. (2012).

Theodoridis (2011), following Canova (1994, 2007), notes that many DSGE economistis prefer to use partial information estimators rather than full information ones, like maximum likelihood, because they prefere estimates that minimise the economic distance rather than statistical fit criteria. Because structual IRF matching estimates deliver selected DSGE moments-statistic as close as possible to those observed in the data. The economist can assess how well the estimated model replicates these targets allowing the researcher to identify possible modelling weaknesses and to infer usefull economic conclusions.

The authors use a medium-scale VAR with four lags, ten quarterly series and three structural shocks. The monetary policy shock is identified with a short-run recursive Cholesky restriction following Christiano et al. (2005). The two technology shocks, neutral and capital embodied, are identified instead with long-run restrictions, following Fisher (2006), which builds on Shapiro and Watson (1988). As summarised by Rossi et al. (2012), the restriction imposed by Altig et al. (2011) is that the two technology shocks are the only shocks allowed to affect the level of labour productivity in the long run; among them, only the capital embodied shock is allowed to affect the relative price of investment in the long run. The VAR is identified so that its impulse responses satisfy these restrictions, and the resulting responses are then matched to their model counterparts to estimate the

structural parameters.

Although the model is estimated from the responses to three shocks, the question that motivates Altig et al. (2011) is a monetary one. What the paper sets out to explain is why inflation responds so slowly to a monetary policy shock while firms in the micro data change prices often, and the resolution runs entirely through χ, which is governed by the cost of varying capital utilisation.

That cost is a statement about what happens to a firm's marginal cost when demand for its output moves while its capital stock is fixed, and a monetary policy shock is exactly that experiment. It shifts demand within the quarter, technology is unchanged, and the capital stock is predetermined, so the joint response of utilisation, hours and inflation is close to a direct reading of Capacity utilisation cost σ_a.

Capacity utilisation is itself one of the ten series in the system, so this is not an indirect inference. A technology shock is a poorer experiment for the same parameter, because it moves the production function and the desired quantity of capital services at once, and what the utilisation response reveals is then a mixture of the two.

This is not to say that the technology shocks are decorative. The DSGE carries three structural disturbances and the estimator uses all of them: they carry information about the investment adjustment cost and about the parameters governing the propagation of supply-side disturbances, and without them several parameters would be poorly pinned down. But they are not where the micro–macro conflict is settled. It is settled by σ_a, and σ_a is read off the response to a monetary policy shock.

The analysis that follows in this work therefore centres on the monetary policy shock. If σ_a turns out to be sensitive to the VAR's specification, so is the reconciliation Altig et al. (2011) propose, and the sensitivity comes from the monetary policy block, the part of the model this paper is really about.

The two technology shocks are nonetheless reported and discussed throughout, rather than set aside. They are identified in a different way, through restrictions on their effects in the long run rather than on their behaviour within the quarter, and that difference will prove informative in its own right.

2.4 SVAR and IRF minimum distance

The SVAR can be recovered by imposing structure on the reduced form (RF),

$$B_0^{-1}B_0Y_t = B_0^{-1}B_1Y_{t-1} + \cdots + B_0^{-1}B_pY_{t-p} + B_0^{-1}w_t, \tag{3}$$

which matches the RF-VAR term by term once $A_i = B_0^{-1} B_i$ for each lag i and

$$u_t = B_0^{-1} w_t. \tag{4}$$

Equation (4) is the bridge between the two representations: the reduced-form residuals are a linear combination of the structural shocks. Taking variances on both sides,

$$\Sigma_u = B_0^{-1} \Sigma_w \left(B_0^{-1}\right)'. \tag{5}$$

Σ_w is the variance-covariance matrix of the structural shocks w_t. These are assumed mutually orthogonal and normalised to unit variance, so $\Sigma_w = I_K$ by construction. This is a normalisation, not an assumption about the economy: what is estimated is not the scale of each shock, but how a given shock propagates through the system, which is entirely encoded in B_0^{-1}.

Y_t is the ten-dimensional vector of series used by Altig et al. (2011),

$$Y_t = \begin{pmatrix} \Delta \ln(\text{relative price of investment}_t) \\ \Delta \ln(\text{GDP}_t/\text{Hours}_t) \\ \Delta \ln(\text{GDP deflator}_t) \\ \text{Capacity utilisation}_t \\ \ln(\text{Hours}_t) \\ \ln(\text{GDP}_t/\text{Hours}_t) - \ln(W_t/P_t) \\ \ln(C_t/\text{GDP}_t) \\ \ln(I_t/\text{GDP}_t) \\ \text{Federal funds rate}_t \\ \ln(\text{GDP deflator}_t) + \ln(\text{GDP}_t) - \ln(\text{MZM}_t) \end{pmatrix}. \tag{6}$$

All the variables are quarterly series from the 1959Q2 - 2001Q4 period, more details on data are described in Appendix A.

Identification combines a short-run and a long-run restriction. The monetary policy shock is identified recursively, following Christiano et al. (2005): the interest rate reacts to lagged variables and to a subset of fast-moving contemporaneous variables, but the shock is assumed orthogonal to that information set, which amounts to zero restrictions on B_0. The two technology shocks are identified through long-run restrictions as in Fisher (2006): both neutral and capital-embodied technology shocks are the only shocks allowed to affect the long-run level of labour productivity, and the capital-embodied shock alone is allowed to affect the long-run relative price of investment. The full derivation is in Appendix B.

These restrictions identify three shocks out of ten. IRFME matches the IRFs of the three identified shocks, not on those not identified.

2.5 Estimating the Parameters of the Equilibrium Model

Using Altig et al. (2011) notation, let ζ denote the strucutal model parameters that are aimed to be estimated, and $\Psi(\zeta)$ being the mapping from the structural parameters ζ to the model IRFs. The SVAR IRFs estimates are denoted as $\hat{\Psi}$.

The IRFME then can be written as:

$$\hat{\zeta} = \arg\min_{\zeta} (\hat{\Psi} - \Psi(\zeta))' V^{-1} (\hat{\Psi} - \Psi(\zeta)) \tag{7}$$

The IRF matching is done horizon by horizon: the estimator compares the VAR point estimate of each impulse response at each of the first h horizons; being $h = 20$ in Altig et al. (2011) work, excluding those set to zero by identification, against the model-implied counterpart. Here, V is a diagonal matrix composed by the sample variances along the diagonal $\hat{\Psi}$. Those variances, compose the condifence intervals around the point estimates of the SVAR IRFs. With this choice of V, the estimated model's structural parameters $\hat{\zeta}$ are the value of ζ that ensures that the $\hat{\Psi}$ DSGE's IRFs lies as much as possible inside the VAR IRFs' confidence intervals. Standard errors of the DSGE's structural parameters $\hat{\zeta}$ are computed using the Deltha method, more detail in the Appendix E

The minimum distance estimator relies on the ability of the VAR to estimate the IRFs. *"if the VAR is misspecified, the specification errors will be computed at each horizon"* point the attention on misspecification concerns in general VAR practice (Ramey, 2016, p. 83). This is clear to the author which claim *"The quality of our strategy (IRF matching) depends on the ability of identified VARs to generate reliable estimates of the dynamic response of economic variables to shocks"* (Altig et al., 2011, p. 3, fn. 10).

3 Truncation Bias and the Case for Long-Lag VARs

Many DSGE's log linearised solution can be mapped in State Space representation, and then, under some assumption can be re-written as VAR(∞) Fernández-Villaverde et al. (2007). Standard lag choice by SVAR practitioner is to use forecasting information criteria (IC) such as Akaike (AIC) or Hannan-Quinn (HQ) and Schwartz (SW), but the most common choice is use conventions; that is, 4 lags for quarterly data and 12 for monthly data.

If a DSGE is estimated using an IRF matching tecnique, using a short lag VAR it might be problematic as the DSGE is estimated by a biased by truncation VAR. VARs are typically small systems with few lags often interpreted as models of the entire economy, IRFs are highly sensitive to variable and lag choice Barnichon et al. (2026); De Graeve and Westermark (2025).

Truncation Bias is a type of misspecification that arises when the VAR lag order is

shorter respect its DGP. This type of misspecification might be a potential problem for structual inference.

3.1 Illustration of the problem: Truncation Bias

Following De Graeve and Westermark (2025), consider a data generating process in which y_t depends on two exogenous variables,

$$y_t = x_{1t}\beta_1 + x_{2t}\beta_2 + \epsilon_t. \tag{8}$$

Estimating the relationship while omitting x_{2t},

$$y_t = x_{1t}b_1 + e_t, \tag{9}$$

Equeation 9 delivers a biased point estimate unless the regressors are orthogonal, that is $\mathbb{E}(b_1) \neq \beta_1$, and an upward-biased residual variance, $s^2 > \sigma^2$. Both distortions matter here, and the second is the one usually forgotten.

The same logic carries over to a VAR. Let

$$Y_t = B_1 Y_{t-1} + \cdots + B_p Y_{t-p} + u_t, \qquad \mathbb{E}[u_t u_t'] = \Sigma_u, \tag{10}$$

with

$$B(L) = B_1 L + \cdots + B_p L^p \tag{11}$$

where $p \ll p^*$, where p^* denotes the true lag order.

Truncation biases both the autoregressive coefficients $B(L)$ and the reduced-form covariance matrix Σ_u. Since impulse responses are a function of both, they inherit the bias.

$$\text{IRF} = f\big(B(L), \Sigma_u\big), \tag{12}$$

Written in this way, however, the statement treats Σ_u as though it were itself the object that matters for structural analysis. It is not. The reduced form covariance is only ever used through a further mapping into structural shocks, and whether that mapping inherits the bias in $B(L)$ turns out to depend entirely on how the shock of interest is identified.

3.2 Identification bias

Ravenna (2007) finds two channels through which lag truncation operates. The first being *pure truncation* bias the one described in the section above. The second he calls it *identification bias*, and works trought the fact that impulse responses are not read off the

reduced form alone, they also require the map from reduced form innovations to structural shocks, and that map has to be estimated as well.

Ravenna's identification bias arises when the identifying restrictions are correct, so it does not originate in mistaken identification assumptions, and it arises in population, so it is not an artefact of small samples.

For this reason *truncation on identification bias* could be a more accurate label. Whenever the estimated autoregressive coefficients enter its computation, a biased $\hat{B}(L)$ delivers a biased $\hat{B}_0$, and the object the researcher labels a structural shock is not the shock the identifying assumption defines.

Whether the second channel, is present therefore reduces to a single question: do the estimated autoregressive coefficients enter the computation of the identifying map?

Under a recursive short run scheme they do not. The identifying restrictions are zero restrictions on contemporaneous relations, so B_0 is recovered from $\hat{\Sigma}_u$ alone, and the lag polynomial plays no part in it. This does not make the recursive scheme immune to pure truncation, since $\hat{\Sigma}_u$ is itself inflated by the omitted dynamics and the propagation is distorted as it is for everyone else. The short run Cholesky scheme is not affected to the second channel, truncation on identification, because the restriction is a statement about the contemporaneous covariance of whatever innovations the estimated VAR produces, and it retains that meaning at any lag order.

Under a long run scheme they do. Long run identification is known to be delicate for several reasons, of which truncation is only one (Faust and Leeper, 1997; Erceg et al., 2005; Chari et al., 2008; Christiano et al., 2007). The mechanism at work here is clearest in the formulation of Blanchard and Quah (1989), where the identifying map is obtained by factorising the long run covariance matrix

$$[I - B(1)]^{-1} \Sigma_u \left\{[I - B(1)]^{-1}\right\}', \qquad B(1) = \sum_{j=1}^{p} B_j, \tag{13}$$

so that the entire lag polynomial evaluated at unity enters the identification directly. A VAR(p) does not estimate that sum consistently when the true representation is of infinite order, and the restriction is therefore imposed on the wrong object.

Altig et al. (2011) impose the same kind of restriction by a different route, following Fisher (2006): instead of rotating an estimated matrix after the fact, they build the restriction into the estimating equation and read the identified shock off as a regression residual (mechanics in Appendix B). Either way, the restriction binds on the sum of the lag polynomial the VAR actually estimates, which runs only to p, not on the sum over the

true $p^* > p$ lags it is meant to capture. The restriction is anchored to the wrong object, and the shock it delivers is the wrong linear combination of reduced form innovations.

This has a consequence for where the bias may appear. A long run restriction says nothing about any finite horizon, impact included. It fixes the cumulated effect and leaves every response along the way to be implied by the estimated dynamics. The long run restriction is therefore not a constraint on the response at a given horizon: it is the rule by which the shock is isolated from the reduced form innovations, and it is on that rule that truncation acts. Applied to a truncated polynomial, the identifying restriction can isolate a different combination from the intended one. If the shock itself turns out to be a different one, the bias is no longer bound to accumulate with the horizon: it could already be present at the first period, unlike pure truncation bias which builds up over time.

The two channels leave different traces: *pure truncation* bias distorts propagation, so its mark accumulates with the horizon, while *truncation on identification* bias distorts the shock, so its trace could be visible already on impact.

The VAR of Altig et al. (2011) carries both kinds of restriction at once, which makes it an unusually clean setting in which to look for both the cannels.

In essence, both the channels in which truncation bias potentially operates can be closed or mitigated by simply increasing the lag lenght of the VAR. Applied practice has largely resisted that response. Even where long-lag VARs are known to have good asymptotic properties Plagborg-Møller and Wolf (2021), they are widely treated as unworkable in finite samples due to high variance concerns: more lags mean more parameters, fewer degrees of freedom, and confidence intervals presumed to widen accordingly. That presumption is part of why a recent literature prefers Local Projections (LP) to VARs outright Li et al. (2024); Montiel-Olea et al. (2025).

De Graeve and Westermark (2025) explore the trivial solution to short lag VARs, that is, dramatic increase in the lag lenght.

3.3 Long-Lag VARs

Their central point is that standard intuition is just a portion of the story, *"in face of misspecification due to lag truncation, increasing the lag length can actually reduce unceirtanty"* (De Graeve and Westermark, 2025, p. 2).

As the dynamic structure increases, lag truncation decrease, consequently misspecification reduces. Misspecification reduction has a double effect, the first being the well-known bias reduction, while the second is the variance reduction that works against the impreci-

sion coming from the increased parametrisation. This trade off it applies to all truncated VARs, no matter what the identification strategy is.

To reveal Long-Lag VARs properties De Graeve and Westermark (2025) generate data, with a MonteCarlo, of realistic samples size from three DSGE, to estimate VARs of different lag length. The three DSGE are Chari et al. (2008) hencefore, Christiano et al. (2007) and Smets and Wouters (2007).

This MonteCarlo simulation is conducted under the assumptions that the identification restriction is correct. And the DSGE are fundamental in the sense of Fernández-Villaverde et al. (2007), so invertivility is never a problem. Under these ipotesis these models have an exact VAR(∞) representation.

With the MonteCarlo they generate 1000 datasets of realistic macro sample size for quarterly data, $T = 200$, and they estimate the bivariate VARs [5] representations of the DSGE choosing different increasing the lag choice. When comparing VARs of different lag-lenght, they restrict the sample of effective observations as $T = 170$ ensuting that lag inizialization does not affect sample size. Unceirtanty bands are computed as in (Sims and Zha, 1999; Canova, 2007; Uhlig, 2005), in this sense it is used a the Normal-Inverse Wishart prior, which have good aproximation to frequentist bootstapped intervals. It does not weigth differently short vs. long lags as would do other prios like the Minnesota one.

3.4 Long-Lag Benefits

De Graeve and Westermark (2025) analyse how the truncated VARs of the three DSGE with VAR(∞) representation behave. The results are much in favor for long-lag VARs. The variance-reducing effect of lower misspecification often dominates the penalty of increased parametrisation. In a nutshell, Long-Lag VARs, reduce truncation bias, have more precise inference, reduced Mean Squared Error (MSE) and better coverage.

Short-lag VARs have maximum unceirtanty for horizons where unceirtanty is not mechanically low. For longer horizons, short-lag VARs have minimum CI width, this because a VAR(p) cannot propagate much beyond horizon p, VARs with lag length extrapolate impulse responses for horizons $h > p$ from the first p autocovariances Baumeister (2025). Therefore unceirtanty cannot propagate much beyond that horizon either.

Unceirtany does not explode with the lag lenght increase; instead for VARs with very long lags, confidence bands are roughly in the same ballpark as those of short-lag VARs.

Long-Lag VARs have comparable coverage rates and comparable MSE than short-lag VARs. combined with the tendency to produce smaller biases, long-lag VARs have favorable properties compared to standard short-lag VARs.

[5] For Smets and Wouters (2007), they reduce the number of variables from 7 to 2 to maitanin consistency with the other two exercises. They keep GDP-growth and interest rates.

De Graeve and Westermark (2025) leverage to the above mentioned MonteCarlo simulation proposing an exercise that decompose the unceirtanty effects creating hypotetical IRFs that are composed by long-lag estimates of the point estimates, while using short-lag estimates of variance covariance matrix obtaining an $IRF = f(B_{30}(L), \Sigma_4)$. Such IRFs can be interpreted as isolating the effect of increased parametrisation.

The standard intuition whould suggest that long-lag length makes the entire distribution shift outward, through the additional unceirtanty created by the strong increase of parameters. In their exercise find that the increased parametrisation does not invariably increase unceirtanty us at odds both with standard intuition (less degrees of freedom) and with the well known omittet variable result.

Standard intuition on increased parametrisation is partially correct and obviously part of the story, but De Graeve and Westermark (2025) finds that misspecification reduction tends to have substantial variance effects.

Given that macroeconomist almost never have the luxury of knowing the true DGP as in the just reviewed the MonteCarlo exercise. Baumeister (2025) (in the discussion of Montiel-Olea et al. (2025)) and also De Graeve and Westermark (2025) ask if long-lag VARs are a viable methodology in sample data revisiting some well known empirical studies.

Qua creare ponte di collegamento tra discussione in long-lag VARs quando non si conosce il DGP: meglio andare lunghi! sopratutto se la domanda economica richiede è di lungo periodo, o ad ogni modo, bisogna dimostrare di essere robusti a long lag VARs given the what it has been studied respet the trustfullness of point estimates of IRFs related to p and h: here there is some to be done: recall that, given that even if the economic quesiton does not relates to long run stuff, IRFME matches VAR IRFs' up to h periods, so in ligth of above, even if we are estimating a DSGE that aims to study short run stuff, if it is done with IRFME we need to take care about h!. Then some examles of pure structural questions with VARs affected by long lag estimation and in particualr the ones with econ questions which are linked to the long run. usare is seguente The misspecification concerns of short lag VARs which potentially suffer from truncation bias naturally links with the IRF matching estimator. Its role is to minimise the distance of the sample IRFs with the theoretical ones, in order to estimate the model's strucural parameters. Sample IRFs are the objective that the estimator wants to mimic, but if they are misspecified, the estimator is targeting the strucutal parameters to an object that is biased, consequently obtaining potentially distorted estimates of the strucutal parameters.
De Graeve and Westermark (2025) also revisits Galí (1999) and Chang and Hong (2006) (with uninformative priors) and Miranda-Agrippino and Ricco (2021). While Baumeister (2025) re estimate Jordà et al. (2024) BVAR with 60 monthly lags and complitely break down their result on long run effects of monetary policy.

Antolín-Díaz and Surico (2025) use a very long-lag VAR to asses long-run effects

question, they activly are using VAR for making strucutral inference on h periods while using $p = h$ lags.

3.5 Model size

The bias-variance trade-off documented by De Graeve and Westermark (2025) applies to all untruncated reduced form VARs. It does not depend on the number variables in the VAR, nor in the identifying assuption or wheater the researcher is using a Bayesian or Frequentist method.

The Frequentist researcher migth bump into a degrees of freedom problem. An n-variable VAR(p) has $n + n^2p$ coefficients, and $\frac{n(n+1)}{2}$ elements in the variance covariance matrix. As the lag lenght grows, the number of parameters to estiamte will rapidely increase, and the higher is the number of variables in the VAR, the faster will happen. At some point the number of parameters to estimate will be greater than the number of available data points. It is shown by De Graeve and Westermark (2025) that the degrees of freedom limit is far from being reached by small scale VAR with one year of lags. In this work replicating Altig et al. (2005) we stand in the Frequentist side, so I won't talk regularization tecquicues or how easy is to implement long-lag VARs in Bayesian as they inherently avoid over-parametrisation.

Given the dimension of VAR in analysis, I am going to show that even being close to the limit, it is possible to estimate long(er)-lag VARs in the Frequentist approach.

3.6 Chosing the Lag Lenght

The one lessons that comes from De Graeve and Westermark (2025); Baumeister (2025) on lag choice selections is that, for making structural inference with VARs, the lag lenght p shoul be chosen consistenly with the horizon of interest of the research question. GIA' DETTO, PENSARE COME RIFORMULARE O EVOLVERE QUA: For a VAR(p) inference is largely uninformative at horizons $h > p$ since the VAR(p) cannot propagate much beyond the horizon p. Because it implies that lag truncation leads to unceirtanty being meccanicaly underestimated at those longer horizons. *Coupled with potentially significant truncation bias, the short-lag VAR researcher runs the risk of becoming relatively certain about the wrong point estimate. Hence, if the research question centers on long horizon effects the VAR better include lags that extend that far."* (De Graeve and Westermark, 2025, p. 2).

What emerges from De Graeve and Westermark (2025) leverage that the results that came from implementing enriched lag struture are solely to take into consideration for structural inference works. None fo the results imply that long-lag VARs are advised to use for forecasting matters. In these type of exercises parsimony hedges against overfitting, that is a matter of poor performance in forecasting unseen data. But if one wants to draw

structural concusions, IRFs analysis or use VARs to estimate DSGEs with a IRFME, then, misspecification concerns are essential.

In these sense, if structural inference is the objective of the work, it may be ill-advised to trust lag-selection criteria with purely focus on forecasting and parsimony as AIC or HQ and SC. These lag selection criteria are not designed for the pourpose of dynamic causal analysis, parsimony introduces truncation bias Barnichon et al. (2026); Baumeister (2025). Connecting to Altig et al. (2005), their IC suggest to use 1 ore 2 lags, the half of the choice taken by the authors. NOT THE PLACE FOR THIS SENTENCE, QUESTO VA MESSO IN EMPIRICAL RESULTS Both Baumeister (2025); De Graeve and Westermark (2025) points on the necessarity to leave forecasting IC for lag selection when structural inference is the aim of the anaylisis, and González-Casasús and Schorfheide (2025) are proposing a criteria that acknowledge the model will be used for structual inference.

But it is not necessary to wait for state of the art IC to implement long lag VARs. *"A simplier heuristic approach would be to plot the structural moments central to the research question against lag length. If the structural conclusion dramatically change as the lag lenght increases, that is clearly indicating that truncation is important to the question at hand"* (De Graeve and Westermark, 2025, p. 12).

"The presumption that adding more lags necessarly leads to prohibitively large unceirtanty is hereby disproven" (De Graeve and Westermark, 2025, p. 12).

qua devo unire De Graeve and Westermark (2025); Baumeister (2025), e se riesco anche Ludwig (2026), ed infine anche solite cit rispetto a truncation bias that affects IRFs at every horizon, including those at h ¡ p. Devo riuscire a chiudere con qualcosa tipo : in IRF matching we are matching h horizons of IRFs, so even if the research question is about short effects, we need to take in mind that we are matching h horizons, so we need h consistent estiamtes of IRFs, therefore p should be consistent with h

4 Empirical Results

In this section I present results from enriched dynamic structure of Altig et al. (2005) VAR used to estimate their DSGE parameters. Doing this exercise, I extend the work of De Graeve and Westermark (2025) in three different aspects.

First, in MonteCarlo laboratory that De Graeve and Westermark (2025) have created to evaluate the three DSGE that have choose to use to generate data, all respect an important assumption, Fernández-Villaverde et al. (2007) Poor Mans condition. All the DSGE admits a VAR(∞) representation, and because of this they can evaluate the performance of truncated VARs, which are full informative. Are bivariate, have two variables and two shock. The VAR used in Altig et al. (2005, 2011) contronts three shocks with ten observable variables. The system is therefore stocastic singular, there is no exact VAR(∞) representation of the DSGE that the VAR is being used to estimate

for. But this is not a caveat, is a feauture, is one of the reasons why the ACEL used to estimate their model a partial information approach as IRF minimum distance estimator.

The second departure as previously anticipated in the Section 3.3 relates on the dimension of the VAR emploied in this work. Becuase the three DGP processes in De Graeve and Westermark (2025) are two-variable systems, their VAR remains cheapely parametrised even at long lag lenghts, and increasing p costs relatively little in terms of degrees of freedom. The ACEL VAR doess not enjoy that luxury. With $K = 10$ variables, a constant and p lags, each equation carries $10p + 1$ coefficients, and the sample runs from 1959:Q2 to 2001:Q4, that is 171 quarterly observations before lags are consumed. I will show that concerns related to CI explosions migth not accour even in limiting cases.

Three, which is strictly connected to the second, I am using as baseline result the exercises where I am not restricting the sample size to the one used in the longest-lag VAR. The reason is why the results does not vary much, and when they do they do by a small amount and consistenlty with the baseline when the common available data is used. it is shown in Appendix G. The other reason is that, in practice, if a researcher has to decide to use a longer-lag VAR respect the short one, the short VAR wpuld be still estimated with as many data points available. In particular this argument is even more valid when the VAR has a lot of variables or if the available data points is little.

4.1 VARs IRFs

The baseline lag choice of ACEL's VAR is $p = 4$, I re-estimate their DSGE matching it with VARs of lag $p = 8$ and 12, raisig the lags to one up to two and three years. In Appendix G are presented the results when the sample is restricted for all the estimations to the one available to the VAR with twelve lags.

In Figure 2 are plotted the VARs' IRFs to a one standard deviation monetary policy shock easing across different lag structures with 90% bootstrap confidence intervals of the VAR(4). All variables, exept for inflation and the interest rate are expressed in percent terms. Inflation and Fed Funds Rate are expressed in units of percentage points at quarterly rate and annual rate, respectively.

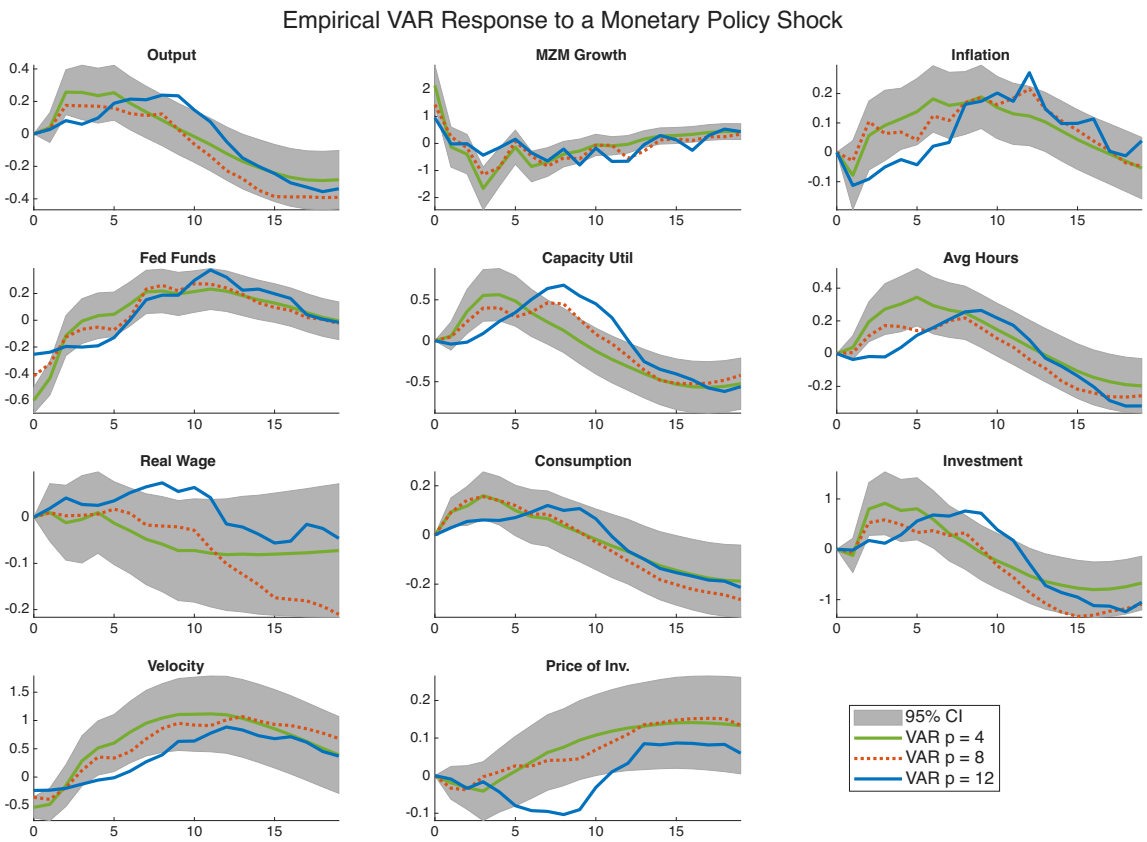


Figure 2: VARs' IRFs to a Montetary Policy shock across different lag struture

The green solid line with grey 90% bootstrap confidence interval are shown the Altig et al. (2005) baseline results from the VAR with four lags. In orange dashed line the IRFs VAR(8), and in blue solid line the VAR(12).

I start desribing Figure 2 from what does not change dramatically. First, the response of money growth rate to a monetary policy shock, consistently with the baseline VAR(4), the effect is completed within one year. Second, there is a liquidity effect, money growth and interest rate move in opposite direction after the shock. Third, velocity co-moves with the interest rate which fall in response to the shock, and then rise. Fourth, price of investment seems not to have an impact, at intermediate horizons have some divergence respect the baseline. In general, all the point estimates across different lags lies inside the confidence intervals of the VAR(4).

Real wage in the VAR(12) have the hump shape respect the baseline which is flat.

Some important responses does not change in the magnitude or sign, nor in the shape, but in the timing. Output, inflation, capacity utilization, average hours, consumption, velocity and investment as the lag choice increase, the timing of the formation of the hump shape, and its peack is shifted in time. On average, the peak effect in the longer-lag VARs occurs one year after respect the baseline results, with the exception of consumption in the VAR(8), response is indistinguable from the baseline.

Fed Funds rate standard deviation shock change from being a 60 basis point at horizon

$h = 0$ which then decrease its impact almost to 0 in two quarters, to a more realistic 25 basis points shock that lasts for five quarters and then goes to 0.

The VAR(12) inflation response need some more attention as it seems to have a puzzeling response in the first year after the easing shock happens. It seems to be very much more intertial respect the baseline, but on the other hand is consistent with all the other response variables that are inertial too. Having in mind Altig et al. (2005) theoretical mechanism we can interpret the inflation response being really slow in the first year after the shock, or even negative, because firms have such strong frictions that allow the portion of optimisers to re-set prices once every more than one year, making inflation and real variables such as output, consumption and average hours being slow to react.

Given the persistence and slow response of the variables in the VAR(12), these can be consisntent with ACEL mechanism if the price stickiness is high implying low implied time duration of prices.

4.2 Unceirtanty around longer lags estimates

Conventional wisdom thinks that increasing lag lenght increase parametrisation, therefore reduce the degrees of freedom making uncertainty around the estimates larger. As well explained by De Graeve and Westermark (2025), this is for sure part of the story. Misspecification lowers the variance of the residuals, and that can offset the imprecision caused by additional parameters. As preaviously mentioned, in a bivariate system there is plenty of room to increase lag lenght without having big concerns relative to overparametrisation. In ACEL VAR the concerns are more than realistic, the parametrisation grows by a factor of three already large relative to the sample. If coinfidence intervals were going to explode anywhere, they would do it here.

They do not. Table 4.2 sumarise the coefficients and degrees of freedom of the exercise. The baseline exercise at $p = 4$, the VAR has 410 coefficients, 167 available data points and 41 coefficient per regression. At $p = 12$ the system carries 1210 coefficients, each equation is estimated on 159 observations with 121 regressors, and only 28 degrees of freedom survive. The degrees of freedom limit, at which the number of regressors per equation k is greater than the number of usable observations T_u, pointed out by De Graeve and Westermark (2025) in this exercise is at $p = 16$.

Tripling the lag lenght relative to the ACEL baseline is therefore not a conservative perturbation, but it is nearly the largest perturbation the data permit. This maks the exercise a demanding test for the central claim in De Graeve and Westermark (2025).

Table 1: Degrees of freedom across lag orders

	VAR $p = 4$	**VAR** $p = 8$	**VAR** $p = 12$
Usable observations T_u	167	163	159
Coefficients per equation k	41	81	121
Degrees of freedom $T_u - k$	126	82	38
Observations per parameter T_u/k	4.07	2.01	1.31
Coefficients, full system	410	810	1,210

Note: T denotes the raw sample size and T_u the usable sample size, $T_u = T - p$; k denotes the number of regressors per equation, $k = 10p + 1$ ($K = 10$ variables, p lags, one constant). The raw sample contains $T = 171$ quarterly observations (1959:Q2–2001:Q4), of which p are consumed as initial conditions, so that $T_u = T - p$. Ordinary least squares requires $k < T_u$, that is $10p + 1 < 171 - p$, that is $p \leq 15$; at $p = 15$ only 5 degrees of freedom per equation remain.

Figure 3 plots the VARs' IRFs to a monetary policy shock at each lag order, togeter with their own 95 percent bootstrap confidence bands.

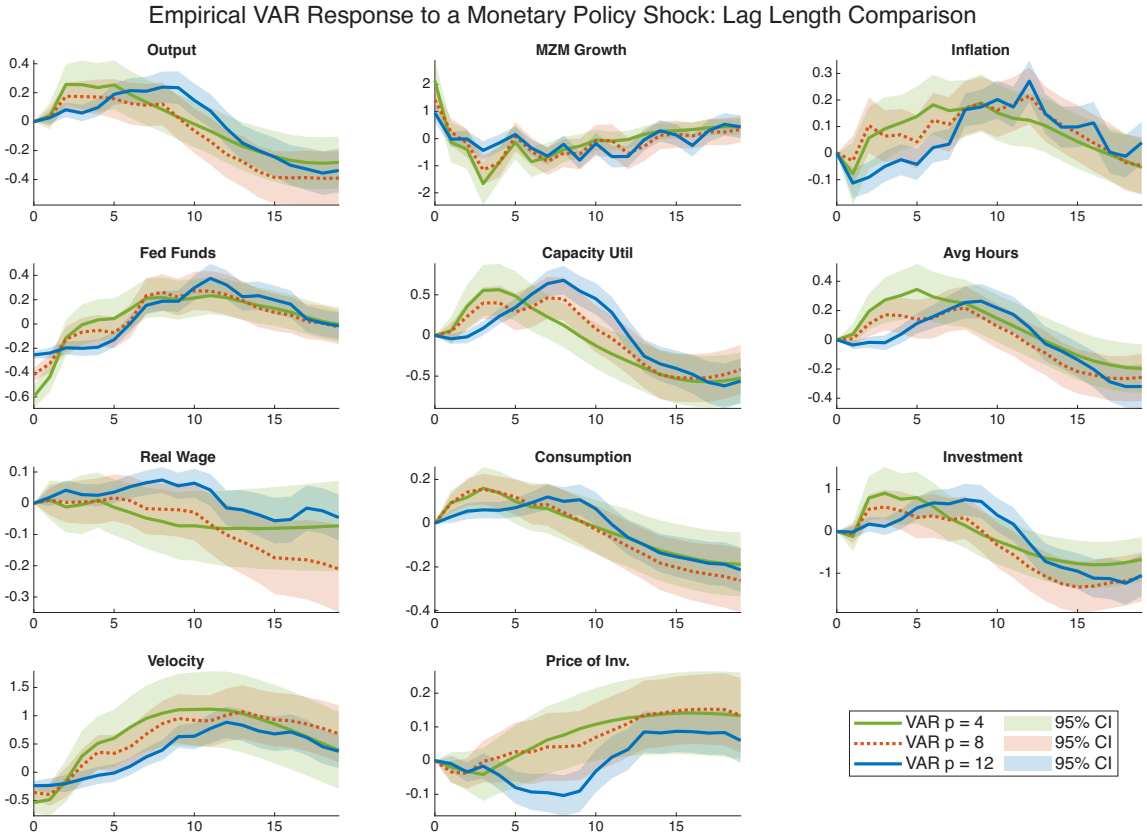


Figure 3: VARs' IRFs to a Montetary Policy shock across different lag struture, 95% CI

The bands at $p = 12$ are not wider that those at $p = 4$ for several variables; inflation, Fed Funds rate, real wage, velocity and relative price of investment, they are visibly narrower despite the increased parametrisation documented in Table 4.2. On average, the width of the CI decrease by XXX Therefore suggesting that the variance reduction mechanism of De Graeve and Westermark (2025) operates even at the parametrisation

fronteer, at least in this work.

The first objection would be that this is simply overfitting: a VAR with 121 regressors per equation fits the sample closely, the estimated residuals shrink, and confidence bands computed from those residuals shrink with them. The objection needs to be taken seriously, because in-sample it cannot be dismissed; reduced misspecification and overfitting have the same in-sample signature, namely a smaller residual variance.

What can be said here, is to point on two direction, misspecification reduction, and bootstrap. Misspecification reduction is at the hearth of De Graeve and Westermark (2025) point, and it is the reason why the confidence bands do not explode with the lag lenght. The latter point is that the bootstrap bands are constructed re-estimating the entire VAR on each artificial sample, so the unceirtanty introduced by the parameters is propagated inot the repoted intervals rather than being conditioned away. If the estimator would have been unstable at $p = 12$, the re-estimated responses would have varied wildy across draws, therefore, the bands would have widened accordingly.

In this sense, bootstrap rules out the mechanical part of the concern, since parameter unceirtanty is not being surpressed. What it cannot be be omitted, is that the residual variance is itself untestated, because the bootstrap resamples from the fitted residuals of the very model under scrunity: NON CONVITNO DI QUESTO, è DI CLAUDE

ACEL choose the lag lenght of $p = 4$ even if the AIC pointed on usign one lag, and HQ and SC suggested to use $p = 2$. The choice was motivated by the fact that the VAR with four lags had a better fit to the data, and that the IRFs were more consistent with the DSGE's theoretical mechanism.

I've checked the IC at longer lags, I find that the AIC criteria points in favour of the VAR(12). The results of this work suggest that the choice of $p = 4$ was not enough to capture the dynamics of the data, and that a longer lag length is needed to obtain more accurate estimates of the DSGE's structural parameters.

5 Structural Implications

5.1 DSGE structural parameters

In this section I analyse what consequences enriched dynamics in the VAR have brought in the DSGE deep parameters.

Table 2: DSGE Structural Parameters Comparison Across Lag Lengths

Parameter	**VAR** $p = 4$	**VAR** $p = 8$	**VAR** $p = 12$
Structural Estimated Frictions			
Price markup (λ_f)	1.0100	1.0100	1.0100
	(-)	(-)	(-)
Wage stickiness (ξ_w)	0.7221***	0.5617***	0.4838***
	(0.1506)	(0.1142)	(0.0757)
Phillips curve slope (γ)	0.0402	0.0402	0.0129***
	(0.0246)	(0.0598)	(0.0042)
Capacity util. cost (σ_a)	1.9938	0.1175	0.0042
	(2.6066)	(0.6299)	(0.0230)
Habit formation (b)	0.7066***	0.6935***	0.8596***
	(0.0463)	(0.0570)	(0.0203)
Inv. adj. cost (S'')	3.2807*	3.0372*	6.8431***
	(1.8874)	(1.5506)	(2.3306)
Money demand elast. (ϵ)	0.8077***	0.7531***	0.8346***
	(0.2111)	(0.1473)	(0.1944)
Implied Price Stickiness			
Homogeneous capital ($\xi_{p,hom}$)	0.8213***	0.8213***	0.8958***
	(0.0480)	(0.0574)	(0.0139)
Firm-specific capital ($\xi_{p,fs}$)	0.3372**	0.6448***	0.8884***
	(0.1505)	(0.1285)	(0.0220)

Note: Standard errors in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The Calvo price parameters standard errors are obtained by Monte Carlo simulation with 1,000,000 draws from the joint distribution of the estimated structural parameters.

5.2 Implied duration and the collapse of the firm-specific mechanism

The Chi parameter

Table 3: Implied Strategic Complementarity Parameter χ Across Lag Lengths

Model Specification	**VAR** $p = 4$	**VAR** $p = 8$	**VAR** $p = 12$
Homogeneous Capital Model ($\chi \equiv 1$)			
χ_{hom}	1.0000	1.0000	1.0000
	(-)	(-)	(-)
Firm-Specific Capital Model			
χ_{fs}	0.0307	0.2030	0.8671***
	(0.0596)	(0.1667)	(0.1698)

Note: Standard errors in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. χ is recovered by inverting $\gamma = \frac{(1-\xi_p)(1-\beta\xi_p)}{\xi_p}\chi$ at the estimated γ and implied ξ_p; $\chi_{hom} \equiv 1$ by construction (Proposition 1(i) in Altig et al., 2011) and carries no standard error. Standard errors for χ_{fs} are computed from the same 1,000,000-draw joint Monte Carlo used for the Calvo parameters.

5.3 Calvo Price Parameters

Table 4: Implied Average Time between Re-Optimization

Model Specification	**VAR** $p = 4$	**VAR** $p = 8$	**VAR** $p = 12$
Homogeneous Capital Model			
Calvo parameter ($\xi_{p,hom}$)	0.82	0.82	0.90
Implied duration (quarters)	**5.6**	**5.6**	**9.6**
Firm-Specific Capital Model			
Calvo parameter ($\xi_{p,fs}$)	0.34	0.64	0.89
Implied duration (quarters)	**1.5**	**2.8**	**9.0**

Note: Implied duration is calculated as $D = 1/(1 - \xi_p)$.

6 Conclusion

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A Data

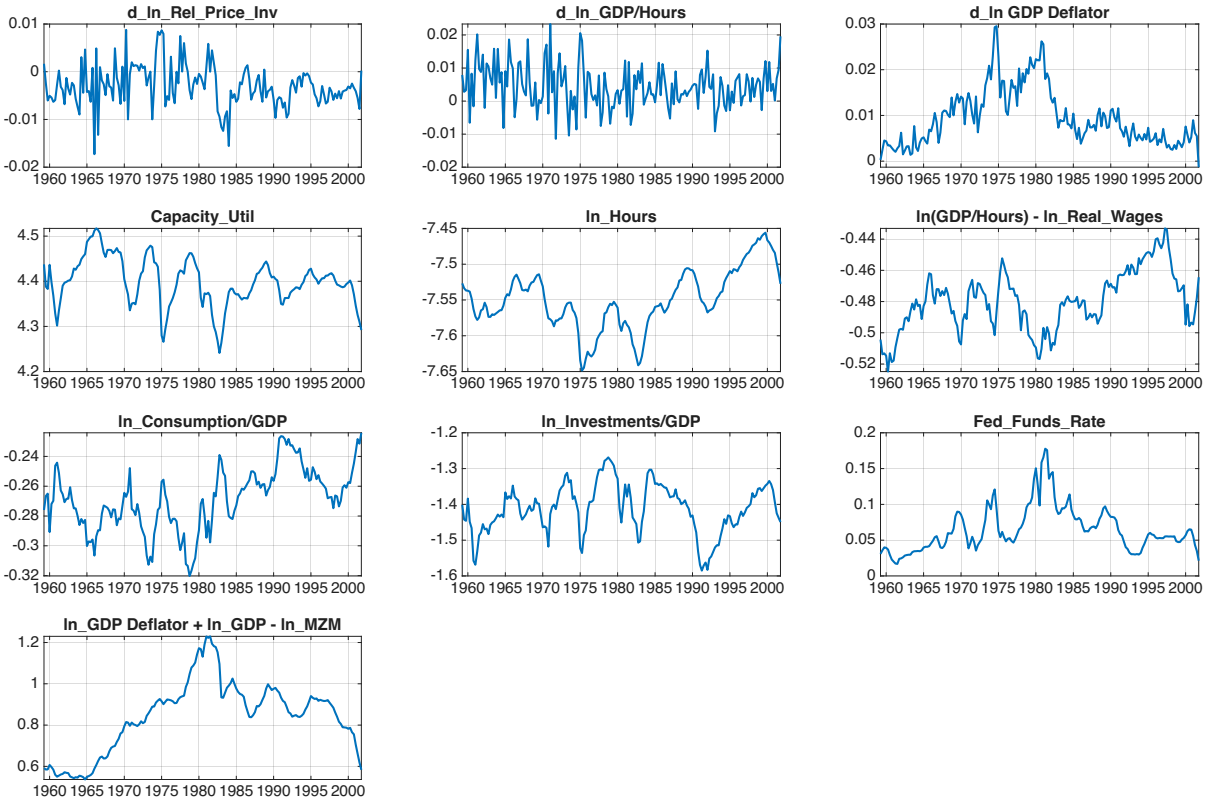


Figure 4: Raw historical series

B Identification strategy

Equation (5) also shows why identification requires restrictions. Σ_u has $K(K+1)/2$ distinct elements, but B_0^{-1} has K^2 free entries, so $K(K-1)/2$ restrictions on B_0 are needed before B_0^{-1} can be recovered from Σ_u. Being numerically sufficient is not enough: the restrictions have to be economically motivated, and it is exactly this requirement that the DSGE is used to satisfy.

C The ACEL Model: Full Derivation

D Additional Residual Analysis

E Delta Method

F Monte Carlo Inference on the Calvo Parameters

G VAR Aligned